Joonsuk Bae

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ALICE PWG-JE Run 3 charged-particle jets — end-to-end pipeline for ALICE's first Run 3 inclusive charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ TeV (preliminary; paper in preparation; EPS-HEP 2025 PWG-JE merge talk on behalf of ALICE). Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.

Education

Ph.D. Candidate, Physics — Sungkyunkwan University 2022.03 – expected 2027.02

Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE.* Defense expected 2026.12; degree conferral 2027.02.

B.S., Physics — Sungkyunkwan University 2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN, Geneva	2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory	2024.02 – present
LAMPS Collaboration, RAON, Korea	2022.03 – 2024.04

Research Experience

Main analyzer, ALICE Run 3 charged-particle jets (PWG-JE) — CERN 2023.04 – present

- End-to-end ownership of the first ALICE Run 3 inclusive charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ TeV — O^2Physics task development, QA, Bayesian/SVD unfolding, full systematic-uncertainty program, and PWG-JE-approved preliminary results. Paper in preparation; presented on behalf of ALICE at EPS-HEP 2025.
- **Tracking-efficiency systematics for Hard Probes 2026 (PWG-JE)** — contributed the pp charged-particle tracking-efficiency systematics at $\sqrt{s} = 13.6$ and 5.36 TeV (p_{T}-binned) for the 2022, 2023, and 2024 periods; the ITS-TPC matching code was reused by a PWG-JE colleague as a starting point for PbPb tracking studies.
- **Trigger-level inefficiency in the new readout** — quantified bunch-crossing and track-matching inefficiencies introduced by the continuous-readout TPC and asynchronous reconstruction (each with its own time-frame); derived signal-loss corrections used in subsequent PWG-JE Run 3 charged-jet analyses.
- **Track p_{T} resolution** — studied data–MC resolution mismatch via covariance-matrix propagation, MC residuals, and V0 mass-resolution methods; tuned the central track momentum-resolution smearing methodology with a p_{T}-dependent profile used in Run 3 charged-jet systematic studies.
- **Cross-section normalization** — established the Run 3 charged-jet normalization based on the TVX visible cross section under continuous readout; authored the PWG-JE cross-section normalization framework (PR #14057, merged) used at particle level by the SKKU Run 3 charged-jet analyses (pp 13.6 TeV and O+O). Provided the normalization factor and QA for the ALICE Run 3 O+O charged-jet measurement led by two MSc co-analyzers in the SKKU group.
- **Event-generator production for model comparison** — managed multi-setup POWHEG-BOX and PYTHIA 8 productions (Monash, colour-reconnection modes, rope hadronization, string shoving) plus HERWIG 7 samples for the Run 3 jet-radius model comparison, with provenance tracking across generator configurations.
- Presented preliminary results at **EPS-HEP 2025** (on behalf of the ALICE Collaboration; PWG-JE merge talk), **Hard Probes 2026** (parallel talk and poster), **Quark Matter 2025**, and **Hard Probes 2024**; regular PWG-JE working-group speaker since 2023.
- Extending the program to semi-inclusive h+jet recoil (PWG-JE Run 3), R-dependent cross-section ratios as a hadronization-vs-perturbative-radiation probe, and groomed substructure (Soft Drop z_g; primary Lund-plane density) in high-multiplicity pp and in the O+O / Ne+Ne data, motivated as a small-system baseline for energy-loss searches.

Detector physicist, ePIC Barrel Imaging Calorimeter (BIC) — Korea-BIC group 2024.02 – present

- Led PMT-module QA prior to test beams — cosmic-ray and source signal characterization, timing-coincidence checks, optical-coupling stability; applied per-module acceptance criteria.
- On-site DAQ shifter at **KEK AF–AR** (2025.03; equalization, prompt analysis; co-author on the corresponding paper in preparation) and lead offline analyst for the **CERN PS T10** (2025.07) campaign of the Pb/scintillating-fiber prototype — energy response, resolution, and linearity (in preparation, 2026).
- Contributed energy- and η-dependent sampling-fraction studies in JANA2 / EIC-recon for the Korea-BIC simulation effort; Geant4 prototype modeling.
- Module assembly and integration — Pb/scintillating-fiber bundle alignment, polishing, PMT/SiPM coupling.

Detector R&D, LAMPS Forward Tracker — RAON, Korea 2022.03 – 2024.04

- Early-stage design and Geant4 simulation of an MPPC–scintillator-plane forward tracking detector; sensor surveys, specifications, and tracking-concept feasibility studies.

Publications

INSPIRE: inspirehep.net/authors/2117232 · ORCID: 0009-0008-4806-8019. ALICE Collaboration co-author since 2022.10.

In preparation

- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* (in preparation). *First ALICE Run 3 inclusive charged-jet publication; thesis paper.* *[main analyzer; analysis, corrections, unfolding, systematic uncertainties; primary author of the internal analysis note and paper draft]*

Submitted (detector)

- Y. Hong, J. Bok, G. An, **J. Bae** *et al.*, "Test-beam performance of the AstroPix silicon sensor for imaging calorimetry," submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00572). Corresponding: Prof. Sanghoon Lim. *[co-author; ePIC BIC silicon-layer test-beam analysis]*
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae** *et al.*, "Beam test of a Pb/SciFi prototype for the Barrel Imaging Calorimeter at the Electron-Ion Collider" — CERN PS T10 (2024.08) campaign; submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482); `arXiv:2604.22647`. *[co-author; module-construction support and PMT/module performance characterization (KPS Spring 2025 poster)]*

ePIC/EIC and related calorimeter R&D

- S. Y. Jang, G. An, **J. Bae** *et al.*, "Performance of dual-readout calorimeters for various absorbers," *Nucl. Instrum. Meth. A* (2026), DOI `10.1016/j.nima.2026.171598`.
- J. K. Adkins *et al.* (ECCE Collaboration), "Design of the ECCE detector for the Electron-Ion Collider," *Nucl. Instrum. Meth. A* **1073** (2025) 170240. *[co-author; ECCE consortium member]*
- G. Cho, S. Kim, **J. Bae** *et al.*, "Beam tests of the copper-based dual-readout calorimeter to measure electromagnetic performance for future e^+e^- colliders," *J. Subatom. Part. Cosmol.* **3** (2025) 100021. *[co-author; module construction and beam-test data taking]*
- ECCE Consortium, "AI-assisted optimization of the ECCE tracking system at the Electron-Ion Collider," *Nucl. Instrum. Meth. A* (2023), DOI `10.1016/j.nima.2022.167748`. *[co-author; ECCE consortium member]*
- ECCE Consortium, "Open heavy-flavor studies for the ECCE detector at the Electron-Ion Collider," `arXiv:2207.10632` (2022). *[co-author; preprint]*

Earlier detector R&D (LAMPS, RAON; peer-reviewed)

- Y. J. Kim, D. S. Ahn, J. K. Ahn *et al.* (LAMPS), "Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON," *J. Korean Phys. Soc.* **87** (2025) 670–682. *[co-author; Time-of-Flight detector maintenance and on-site checks]*
- Y. Bae *et al.* (LAMPS), "Performance test of prototype LAMPS Start Counter using proton beams at 100 MeV," *Nucl. Instrum. Meth. A* (2025), DOI `10.1016/j.nima.2025.170827`. *[co-author; Start-Counter prototype testing]*
- Y. Cheon *et al.* (LAMPS), "Characteristics of the prototype LAMPS TPC for low-energy nuclear collisions at RAON," *Nucl. Instrum. Meth. A* **1066** (2024) 169610. *[co-author; TPC prototype characterization]*
- H. Kim *et al.* (LAMPS), "Performance of the prototype beam drift chamber for LAMPS at RAON with proton and ^{12}C beams," *JINST* **19** (2024) P12008; `arXiv:2412.08662`. *[co-author; forward-tracking R&D and prototype performance studies]*

Software and Computing

ALICE O^{2}Physics — AliceO2Group/O2Physics

- Contributor since 2023: jet-finding and response-matrix tasks, systematic-uncertainty pipelines, particle-level cross-section normalization, and shared track-smearing tools.

Invited and Contributed Talks

Summary since 2022: 3 invited workshops *[external]* · 5 invited collaboration-meeting talks *[internal]* · 8 international conference contributions (one on behalf of the ALICE Collaboration at EPS-HEP 2025; a parallel talk and a poster at Hard Probes 2026) · 9 domestic (KPS) · regular PWG-JE working-group speaker since 2023.

Invited Talks (external workshops, seminars)

- **France–Korea Particle Physics Network (FKPPN) Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **1st Heavy-Ion Meeting (HIM 2025)**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

Collaboration-Meeting and National-Workshop Talks (internal)

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet O^2 MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

Contributed Talks and Posters

- **Hard Probes 2026**, Nashville — *Jet-like dihadron correlations in pO, OO, and Ne–Ne collisions with ALICE [international, talk]* — 2026.06
- **Hard Probes 2026**, Nashville — *Characterizing inclusive charged-particle jet production and fragmentation in pp collisions [international, poster]* — 2026.06
- **EPS-HEP 2025**, Marseille — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE [on behalf of the ALICE Collaboration; PWG-JE merge talk]* — 2025.07
- **INPC 2025**, Daejeon — *Charged-particle jet and jet quenching measurement in pp collisions at $\sqrt{s} = 13.6$ TeV during LHC Run 3 with ALICE [international, talk]* — 2025.05
- **Quark Matter 2025**, Frankfurt — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE [international, poster]* — 2025.04

- **Hard Probes 2024**, Nagasaki — *First measurements of charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE [international, poster]* — 2024.09
- **Quark Matter 2023**, Houston — *Charged-particle multiplicity distribution in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE [international, poster]* — 2023.09
- **ATHIC 2023** (9th Asian Triangle Heavy-Ion Conf.), Incheon — *Dijet studies at the LHC [international, poster]* — 2023.04
- **KPS Spring 2026**, Daejeon *[domestic, talk]* — R*-dependent charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE* — 2026.04
- **KPS Fall 2025**, Daejeon *[domestic, talk]* — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE* — 2025.10
- **KPS Spring 2025** *[domestic, talk]* — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at* $\sqrt{s} = 13.6$ *TeV with ALICE* — 2025.04
- **KPS Spring 2025** *[domestic, poster]* — *Prototype calorimeter QC for the Barrel Imaging Calorimeter at the EIC* — 2025.04
- **KPS Fall 2024** *[domestic, talk]* — *First jet measurement in ALICE Run 3: charged-particle jet production* — 2024.10
- **KPS Spring 2024** *[domestic, talk]* — *Charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV* — 2024.04
- **KPS Fall 2023** *[domestic, talk]* — *First look at charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE Run 3 data* — 2023.10
- **KPS Spring 2023** *[domestic, poster]* — *Dijet studies at the LHC* — 2023.04
- **KPS Fall 2022** *[domestic, poster]* — *Dijet studies with ALICE at the LHC* — 2022.10

Test-beam and Detector Activities

ALICE ITS2 alignment monitoring at CTF level (Run 3)

- Analyzed compressed-time-frame (CTF) data — below the standard AO2D analyzer interface — to provide period-by-period tracking-performance diagnostics.

ePIC Barrel Imaging Calorimeter test beams

- **CERN PS T10** (2025.07) — lead analyst; energy response and resolution.
- **KEK AF–AR** (2025.03) — DAQ, equalization, prompt analysis.
- **CERN PS T10** (2024.08) — first campaign; module performance characterization.
- **Korea-BIC module production** (2024 – 2025) — Pb/scintillating-fiber bundle assembly, polishing, and PMT/SiPM integration; PMT-level QA on the prototype modules used in the 2025 beam-test campaigns.

Service and Professional Activities

- **PWG-JE tracking-efficiency contribution for Hard Probes 2026** — pp 13.6 TeV studies; ITS-TPC matching code reused by a PWG-JE colleague as a starting point for PbPb tracking studies.
- **ALICE Run 3 O+O charged-jet measurement** (PWG-JE) — co-author on the Analysis Note; service contribution: cross-section normalization factor derivation and QA; co-mentor of the two MSc analyzers.
- **ALICE ITS2 alignment monitoring** — service work at CTF level for tracking-performance diagnostics.
- **ePIC Korea-BIC analysis coordination** — analysis liaison between the SKKU group and the Korea-BIC group for beam-test data products.
- **KoALICE national activities** — regular participation since 2022, including invited collaboration-meeting talks.

Awards and Fellowships

- **BK21 FOUR Graduate Scholarship** — Sungkyunkwan University Physics BK21 sponsored unit (Korean Ministry of Education), 2022 – present.

- **Conference travel support** — KoALICE, NRF Korea–CERN program, and SKKU Graduate School; HP 2024 (Nagasaki), QM 2025 (Frankfurt), EPS-HEP 2025 (Marseille).

Teaching and Mentoring

Student mentoring (non-supervisory) — Sungkyunkwan University 2024 – present

- *Wooseok Ham* (M.Sc., co-analyzer on the ALICE Run 3 O+O charged-jet measurement) and *one further MSc co-mentee* (co-analyzer on the same O+O analysis) — mentored on Run 3 jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author with W. Ham on the corresponding PWG-JE Analysis Note.
- *Harim Seong* (undergraduate) — ePIC BIC detector-performance studies using JANA2-based reconstruction; energy response and sampling-fraction extraction.

KoALICE O^2 Tutorial — student co-organizer and instructor — Sungkyunkwan University, Suwon2024.01.22–23

- Co-organized (with Hyungjun Lee) the two-day Run 3 analysis-framework tutorial for the Korean ALICE community — ∼22 participants from SKKU, Inha University, and other Korean institutes; listed as the official Indico contact for the workshop. indico.cern.ch/event/1361235
- Instructor on *Running O^{2}Physics with Hyperloop* (solo) and co-instructor on *Practical PWG-JE tasks* (with H. Lee).

Teaching Assistant, SKKU Physics (2022 – 2024) — *Nuclear Reactions* (graduate), *Quantum Mechanics II*, *General Physics I and Laboratory I*.

Technical Skills

Languages. English (professional working proficiency), Korean (native).
Programming. C++, Python, ROOT, LaTeX, Bash, Git.
HEP frameworks. ALICE O^{2}Physics, Geant4, JANA2 / EIC-recon, FastJet.
Event generators. PYTHIA 8 (Monash, colour-reconnection modes, rope hadronization, string shoving), POWHEG-BOX, HERWIG 7.
Analysis methods. jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.
Detector hardware. PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

References

Available on request.